SMART BLOOD DONOR NETWORK SYSTEM

1. Abstract

Blood donation emergencies are a matter of life and death — yet finding the right donor at the right time remains a slow, manual, and unreliable process. Hospitals call people one by one. Patients' families post in WhatsApp groups. Blood banks run out of stock without warning.

The Smart Blood Donor Network System solves this by bringing donors, hospitals, and blood banks onto one unified digital platform. When a hospital needs blood urgently, the system automatically finds the most suitable nearby donors, ranks them using an AI model, and sends instant alerts — all within seconds.

What makes this system stand out is its intelligence layer: instead of simply listing donors who match a blood group, it predicts which donors are most likely to respond and donate right now, based on their past behaviour. This means hospitals spend less time calling and more time saving lives.

Built as a full-stack web application using the MERN stack (MongoDB, Express, React, Node.js) with a Python-based AI module, this project combines real-time communication, geolocation, machine learning, and analytics into a single platform with genuine real-world impact.

2. Features — Explained Simply

The system has 10 features split into two categories: core features (what any blood donation platform needs) and smart features (what makes this project stand out).

Core Features

1. Donor Registration & Profile

Anyone who wants to donate blood can sign up and create a profile. They enter their blood group, location, and basic health details. Their profile tracks every donation they've made, shows when they're next eligible to donate, and displays badges they've earned (like 'Gold Donor' for 10+ donations). **Donors can also set their availability — for example, marking themselves as 'travelling' so they don't get pinged unnecessarily.**

2. Blood Group Search

Hospitals or anyone in need can search for donors by blood group and distance (e.g., 'Find A+ donors **within 10 km')**. **Results are shown on a map with donor pins**. But unlike a basic search, donors are not shown in random order — they're ranked by a score that factors in distance, availability, and AI prediction (explained in Smart Features below).

3. Emergency Alert System

When a hospital raises an emergency blood request, the system instantly sends push notifications to the top-ranked donors nearby. Donors get an alert on their phone, see the hospital details and distance, and can Accept or Decline with one tap. The hospital sees live updates — '2 donors accepted, 1 en route' — in real time. **If no one responds within 10 minutes, the system automatically widens the search radius and alerts more donors.**

4. Hospital Request Dashboard

Hospitals get their own dashboard to raise blood requests, choose urgency level (Critical / Urgent / Normal), and track every request they've ever made. They can see which donors were notified, who accepted, and how long it took to fulfil the request. This helps hospitals plan better and identify which blood groups are hardest to source.

5. Donation History & Badges

Every confirmed donation is logged in the donor's profile. The system awards badges to motivate donors — First Drop (1st donation), Silver (5+), Gold (10+), Platinum (25+). Donors can also download a PDF donation certificate for each donation, useful for records or college portfolios.

6. Eligibility Reminders

Whole blood donors must wait 90 days between donations. The system tracks each donor's last donation date and automatically sends a reminder via push notification and email 7 days before they become eligible again. This keeps donors engaged and ensures the platform always has a pool of ready donors.

Smart (AI-Powered) Features

7. Donor Availability Prediction (AI)

This is the key AI feature. The system uses a machine learning model **(XGBoost)** trained on donor behaviour data to predict: 'If we send this donor an alert right now, what is the probability they will respond?'

The model looks at factors like:

- How often has this donor responded to past alerts?
- How quickly do they usually respond?
- How far is the hospital from them?
- How many days since their last donation?

The output is a simple probability: High (75%+), Medium (40–75%), or Low (<40%). This is then used to rank donors before alerting them — so the most likely helpers are contacted first.

8. Donor Priority Score (Recommendation Engine)

Rather than showing hospitals a plain list of matching blood group donors, every donor gets a Priority Score out of 100. Think of it like a Swiggy or Zomato ranking — the best option floats to the top.

The score is calculated from five weighted factors:

Factor	Weight	Simple Explanation
AI availability probability	35%	How likely is this donor to say yes right now?
Distance from hospital	25%	Closer donors score higher

Factor	Weight	Simple Explanation
Days since last donation	20%	Longer gap = more eligible and willing
Past response rate	12%	Donors who always help score higher
Total donation count	8%	More experienced donors rank slightly higher

Result: Hospitals see Donor A (Score: 92) before Donor B (Score: 71) — no manual filtering needed.

9. Admin Analytics Dashboard

The admin gets a bird's-eye view of the entire platform through charts and graphs:

- Which blood groups are most in demand? (Pie chart)
- How many donations happened each month? (Line chart)
- Which hospitals raise the most requests? (Table)
- How fast are emergencies being fulfilled on average? (KPI card)
- Are donors coming back after their first donation? (Retention funnel)

This data helps understand platform health and identify gaps — for example, a shortage of O- donors in a particular city.

10. Blood Compatibility Guide + Blood Bank Inventory

A publicly visible compatibility chart shows which blood groups can donate to which (e.g., O- is the universal donor). Blood banks can register on the platform and list their current stock levels by blood group — so hospitals can check bank inventory before even raising a donor alert.

3. Tech Stack — What It Is & Why

The project uses 6 technologies, each chosen for a specific reason:

Technology	Used For	Layer	Why This Choice
React.js	User interface (all 4 dashboards)	Frontend	Fast, component-based, great for real-time UI updates
Node.js + Express	API server, business logic	Backend	JavaScript everywhere; fast, non-blocking I/O
MongoDB Atlas	Store all data (donors, requests, history)	Database	Flexible schema; built-in geospatial query support for location search
Socket.IO	Real-time emergency alerts and live status	Real-time	Two-way live communication between hospital and donor
Python + Flask + XGBoost	AI availability prediction model	ML Service	Python has the best ML libraries; Flask wraps it as a simple API
Google Maps API	**Show donor locations, distances, hospital map**	**Maps**	**Industry standard; easy integration with React**
Firebase FCM	Push notifications to donor devices	Notifications	Free, reliable, works on Android + web browsers
Twilio	SMS + WhatsApp alerts as backup	Messaging	Reaches donors even without the app installed
JWT + bcrypt	Login security, role-based access	Auth	Industry standard; keeps donor/hospital data separate and secure
Recharts	Charts in admin analytics dashboard	Visualization	Lightweight React-native chart library

Tech Stack Workflow

Here's the simple flow of how all technologies work together:

Step	What Happens
1	User opens the React web app in their browser
2	React sends requests to the Node/Express backend via REST API
3	Backend queries MongoDB for donor data and uses geospatial index to find donors by location
4	Backend calls the Python Flask ML service to get availability prediction scores

Step	What Happens
5	Priority scores are calculated and donors are ranked
6	Firebase FCM + Twilio send instant push/SMS notifications to selected donors
7	Socket.IO keeps the hospital dashboard updated live as donors respond
8	All events are logged in MongoDB and reflected in the Admin analytics dashboard

4. Who Uses the System

User	What They Can Do
Donor	Sign up, receive emergency alerts, accept/decline requests, track donation history, earn badges, download certificates
Hospital	Raise blood requests, see ranked donor list, track request status in real-time, confirm donations, view request history
Blood Bank	List blood stock by group, appear on donor search map, accept/reject hospital stock requests
Admin	See all platform analytics, manage users and hospitals, approve new hospital registrations, send broadcast announcements

Simple System Flow

1. Donor registers

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2. Donor profile stored

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3. Hospital raises blood request

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4. System finds matching blood group donors

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5. Distance filter applied

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6. AI predicts donor availability

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7. Priority score calculated

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8. Top-ranked donors notified

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9. Donor accepts/declines

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10. Hospital views responses live

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11. Donation completed

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12. Donation history updated

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13. Badge awarded (if eligible)

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14. Analytics dashboard updated

Identified tables:

Table Name	What it Stores	Purpose
User	Login and account information for all users	Common authentication system for donors, hospitals, blood banks, and admins
Donor	Donor personal details, blood group, location, availability, eligibility	Maintains donor profiles and donor-specific information
Hospital	Hospital information and contact details	Allows hospitals to raise blood requests and manage requests
BloodBank	Blood bank information and registration details	Enables blood banks to participate in the platform
BloodInventory	Blood stock available in blood banks by blood group	Lets hospitals check blood availability before raising requests
BloodRequest	Blood requests raised by hospitals	Central entity that drives the emergency donation process
DonorRequestStatus	Which donor received which request and their response status	Tracks notification, response, and participation history
Donation	Records of completed donations	Maintains donation history and updates donor eligibility
Notification	System-generated notifications and reminders	Handles emergency alerts, eligibility reminders, announcements, etc.
DonorPrediction	AI-generated availability prediction results	Stores probability of donor responding to a request
DonorPriorityScore	Calculated ranking scores for donors	Supports intelligent donor recommendation and ranking